

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Scenario-Based Questions

Code of Conduct:

I N.Tharun Sai (192472382) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Q1. Low Contrast Image

(a) Scenario Interpretation

A grayscale image appears dull because the intensity values of the object and the background occupy a narrow range. As a result, the image lacks brightness variation, making it difficult to distinguish different regions. Low contrast commonly occurs due to poor illumination, incorrect camera exposure, atmospheric conditions such as fog, or limitations of imaging sensors.

(b) Key Issues

- Poor visibility of objects.
- Weak boundaries between object and background.
- Hidden fine details and textures.
- Reduced edge detection performance.
- Inaccurate feature extraction and segmentation.
- Poor visual quality for analysis.

(c) Enhancement Technique

Histogram Equalization is applied.

Algorithm:

1. Compute histogram.
2. Calculate cumulative distribution function (CDF).
3. Normalize the CDF.
4. Map old intensities to new values.
5. Produce the enhanced image with wider intensity distribution.

(d) Justification

Histogram Equalization improves global contrast without requiring prior

knowledge of the image. It enhances visibility of hidden details and improves the performance of later tasks such as segmentation, object detection, and feature extraction. Compared with simple brightness adjustment, it uses the entire grayscale range effectively.

(e) Conclusion

Histogram Equalization is the most suitable enhancement method because it improves contrast, preserves important information, and produces a clearer image.

Q2. Uneven Illumination

(a) Scenario Interpretation

Uneven illumination means different parts of the image receive different amounts of light, producing bright and dark patches. This affects image consistency and reduces analysis accuracy.

(b) Key Issues

- Shadows and highlights.
- Non-uniform brightness.
- Loss of local details.
- Incorrect intensity distribution.
- Reduced segmentation accuracy.
- Feature extraction becomes unreliable.

(c) Enhancement Technique

Contrast Limited Adaptive Histogram Equalization (CLAHE).

Algorithm:

1. Divide image into small tiles.

2. Apply histogram equalization to each tile.
3. Limit contrast amplification.
4. Interpolate neighbouring tiles.
5. Obtain uniformly enhanced image.

(d) Justification

CLAHE improves local contrast while preventing excessive noise amplification. It is more effective than normal Histogram Equalization for images affected by uneven lighting because it processes small regions independently.

(e) Conclusion

CLAHE provides balanced illumination, enhances local details, and preserves natural appearance.

Q3. Salt-and-Pepper Noise

(a) Scenario Interpretation

Salt-and-pepper noise appears as randomly distributed white and black pixels due to transmission errors, faulty sensors, or document scanning. It reduces readability and image quality.

(b) Key Issues

- Random impulse noise.
- Poor readability.
- Text distortion.
- Loss of fine image details.
- Lower OCR and segmentation accuracy.

(c) Filtering Technique

Median Filter.

Algorithm:

1. Select a 3×3 or 5×5 window.
2. Arrange neighbouring values.
3. Compute the median.
4. Replace noisy pixel.
5. Repeat for all pixels.

(d) Justification

The Median Filter removes impulse noise effectively while preserving edges. Mean filtering removes noise but blurs image boundaries, whereas median filtering retains important structures.

(e) Conclusion

Median Filtering is the preferred solution for removing salt-and-pepper noise from scanned documents.

Q4. Blurred Edges

(a) Scenario Interpretation

Noise removal using excessive smoothing reduces high-frequency edge information, causing object boundaries to become blurred.

(b) Key Issues

- Edge information is lost.
- Reduced sharpness.
- Poor boundary detection.

- Lower segmentation quality.
- Decreased visual clarity.

(c) Restoration Technique

Unsharp Masking.

Algorithm:

1. Blur the original image.
2. Subtract blurred image from original.
3. Obtain edge mask.
4. Add edge mask back.
5. Generate sharpened image.

(d) Justification

Unsharp Masking selectively enhances edges without greatly increasing noise. It restores important image structures after smoothing and is widely used in medical, satellite, and photographic imaging.

(e) Conclusion

Unsharp Masking restores edge sharpness while maintaining overall image quality.

Q5. High-Frequency Noise

(a) Scenario Interpretation

High-frequency noise consists of rapid intensity variations that appear as fine grain in satellite images, making geographical features difficult to observe.

(b) Key Issues

- Grainy appearance.
- Reduced feature visibility.
- Poor classification.
- Lower segmentation accuracy.
- Difficulty identifying roads, rivers, and buildings.

(c) Frequency Domain Technique

Gaussian Low-Pass Filter.

Algorithm:

1. Perform FFT.
2. Create Gaussian LPF.
3. Multiply filter with frequency spectrum.
4. Perform inverse FFT.
5. Obtain denoised image.

(d) Justification

Gaussian LPF smoothly suppresses high-frequency noise and avoids ringing artifacts that occur with Ideal Low-Pass Filters. It preserves important image structures better than many alternative frequency filters.

(e) Conclusion

Gaussian Low-Pass Filtering is an effective technique for improving satellite image clarity while preserving useful information.

Overall Summary

Image enhancement techniques improve image quality by increasing contrast, correcting illumination, removing noise, restoring edges, and

enhancing important features. Selecting the appropriate enhancement method based on the type of degradation produces better visual quality and improves the accuracy of image processing tasks such as segmentation, recognition, and analysis.